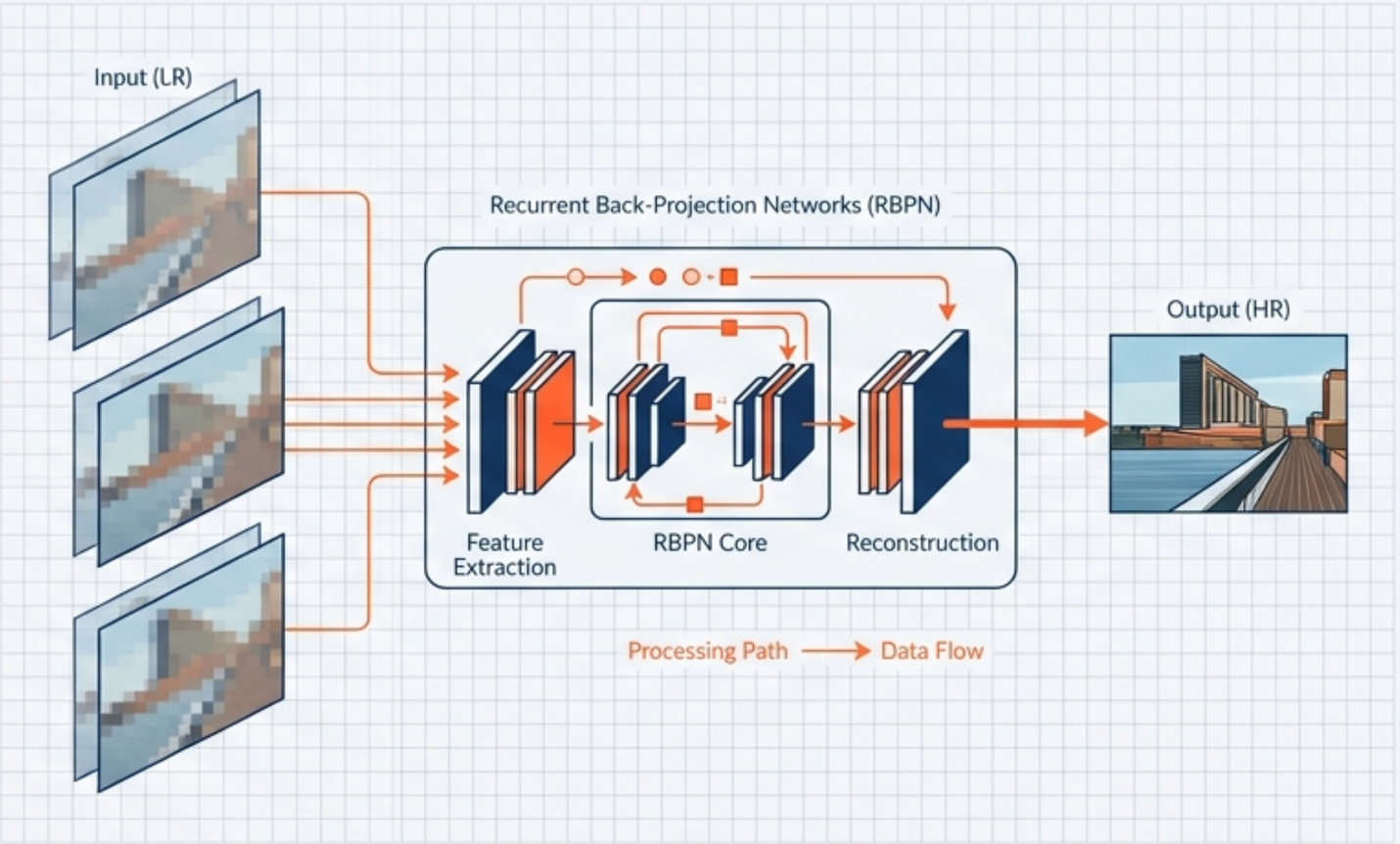
Video Super Resolution Implementation

A Deep Dive into Recurrent Back-Projection Networks (RBPN)



Chapter 22: Mobile Computational Photography

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The Super Resolution Problem

Core Definition

Video Super Resolution (VSR) leverages temporal information to reconstruct **high-resolution** (HR) frames from low-resolution (LR) observations. Unlike Single Image Super Resolution (SISR), VSR recovers **actual lost high-frequency details** by combining data from multiple frames.

Problem Formulation

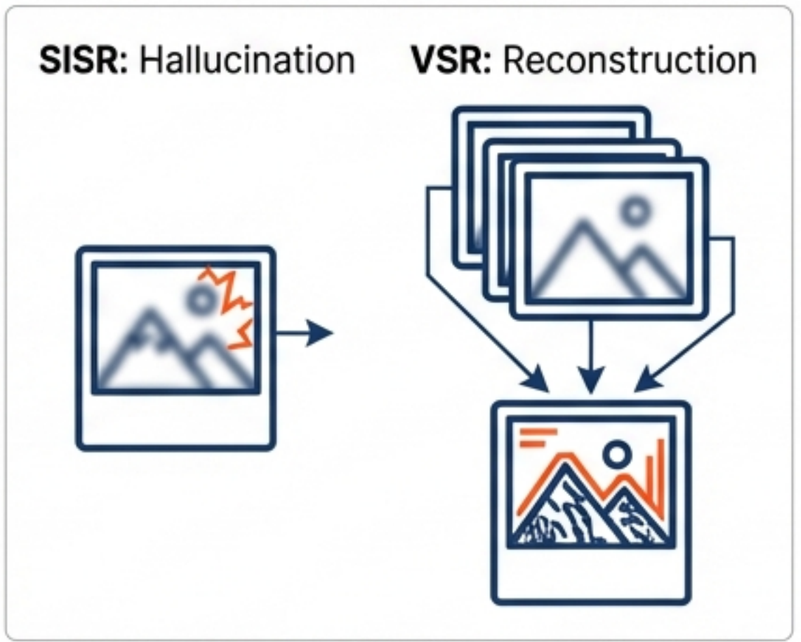
Input: Sequence of Low-Resolution Frames

```
{ I_LR_1, I_LR_2, ..., I_LR_N }
```

Processing Model

Output: High-Resolution Frame at time t

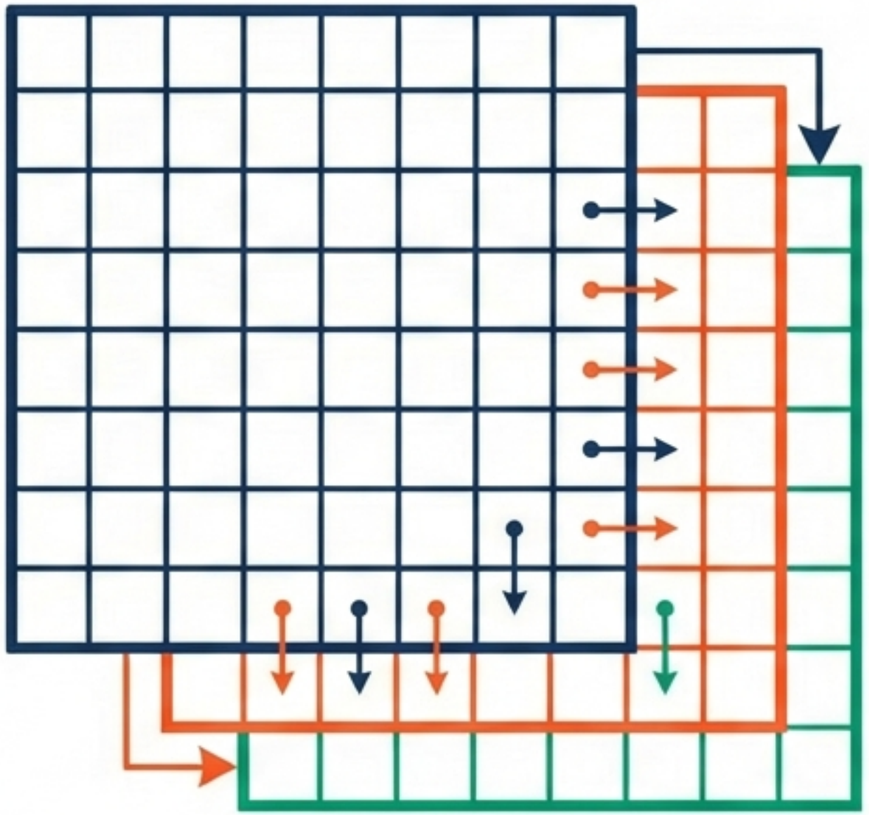
```
I_HR_t
```



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The Advantage of Temporal Data

Sub-pixel Motion



Minute shifts between frames allow the model to sample the scene from slightly different positions,
filling in the gaps of the high-resolution grid.

Key Advantages

Sub-pixel Motion: True detail recovery through multi-view sampling.

Temporal Consistency: Neighboring frames stabilize output and reduce flickering.

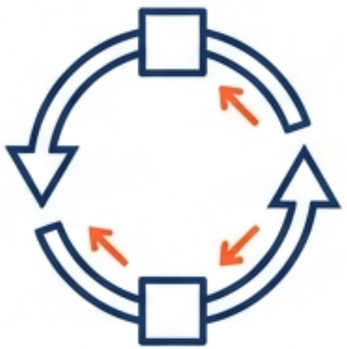
Redundancy: Multiple observations improve robustness against noise.

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The Solution: Recurrent Back-Projection Networks (RBPN)

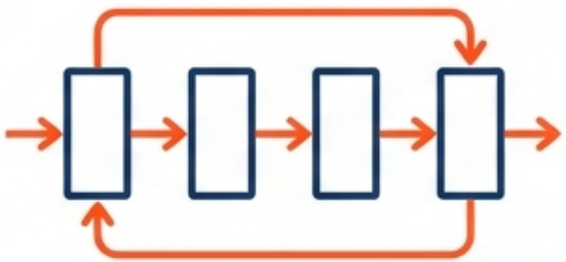
Implementation Architecture (Haris et al., CVPR 2019)

Back-Projection



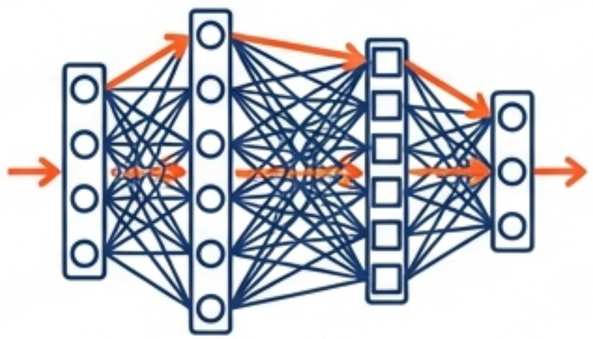
Iterative refinement through error correction. The model compares its output to the input and fixes discrepancies.

Recurrent Architecture



Sequential processing that preserves temporal dependencies across the video stream.

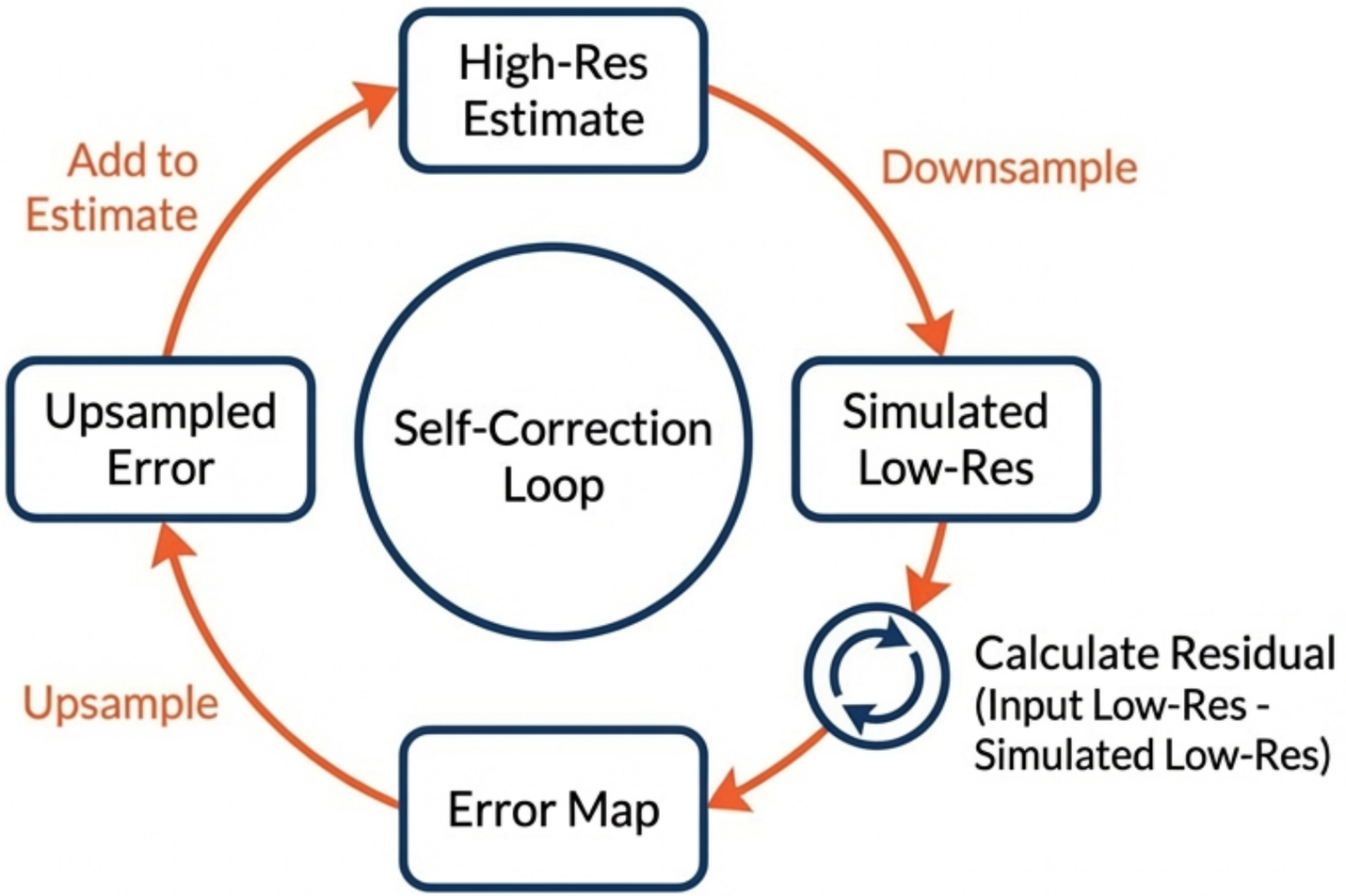
Dense Connections



Facilitates effective information flow, ensuring deep features are preserved for fine detail recovery.

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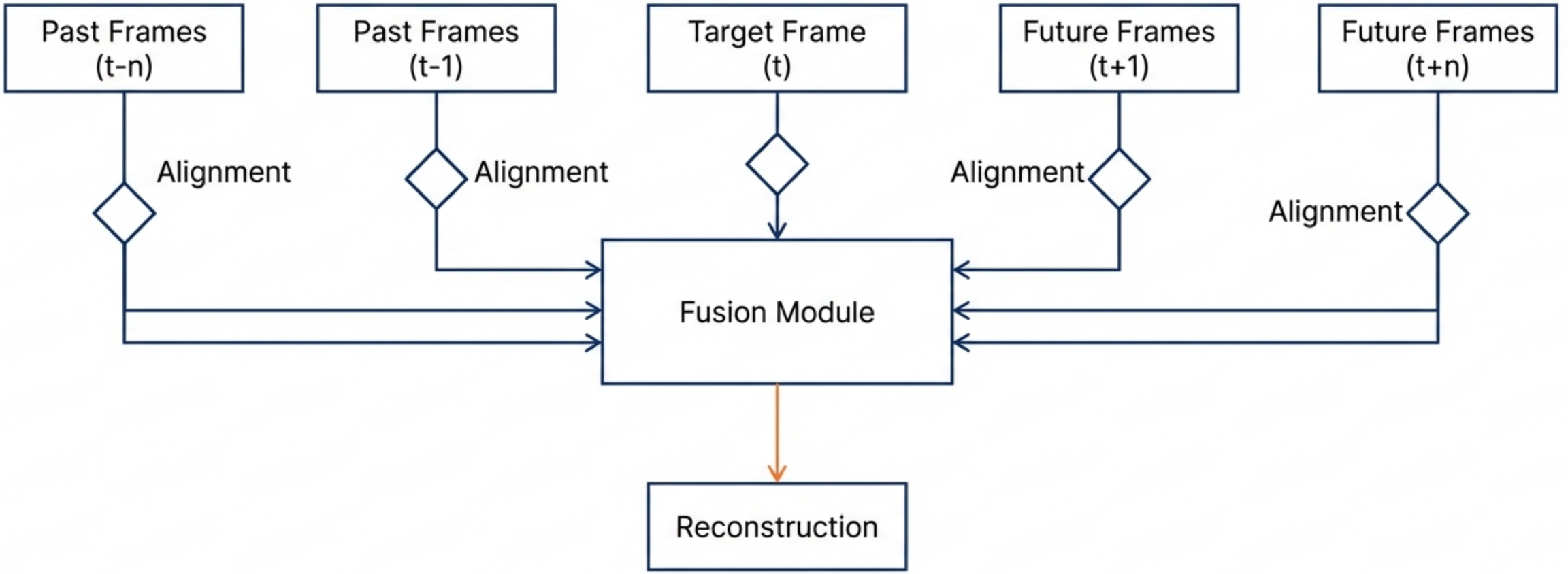
Theory: The Back-Projection Principle



The network 'learns' by constantly checking if its high-resolution guess matches the low-resolution reality when downscaled.

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Theory: Temporal Fusion Strategy



- **Feature Extraction**: Processing each frame in the window independently.
- **Temporal Alignment:** Adjusting features to account for object motion.
  - **Fusion**: Merging features from past and future to support the current frame.

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Implementation: Environment Setup

Prerequisites

- Python 3.8+
- PyTorch ≥ 1.9.0
- OpenCV, NumPy, Scikit-image

Repository Source

https://github.com/JeffRen1977/super_resolution

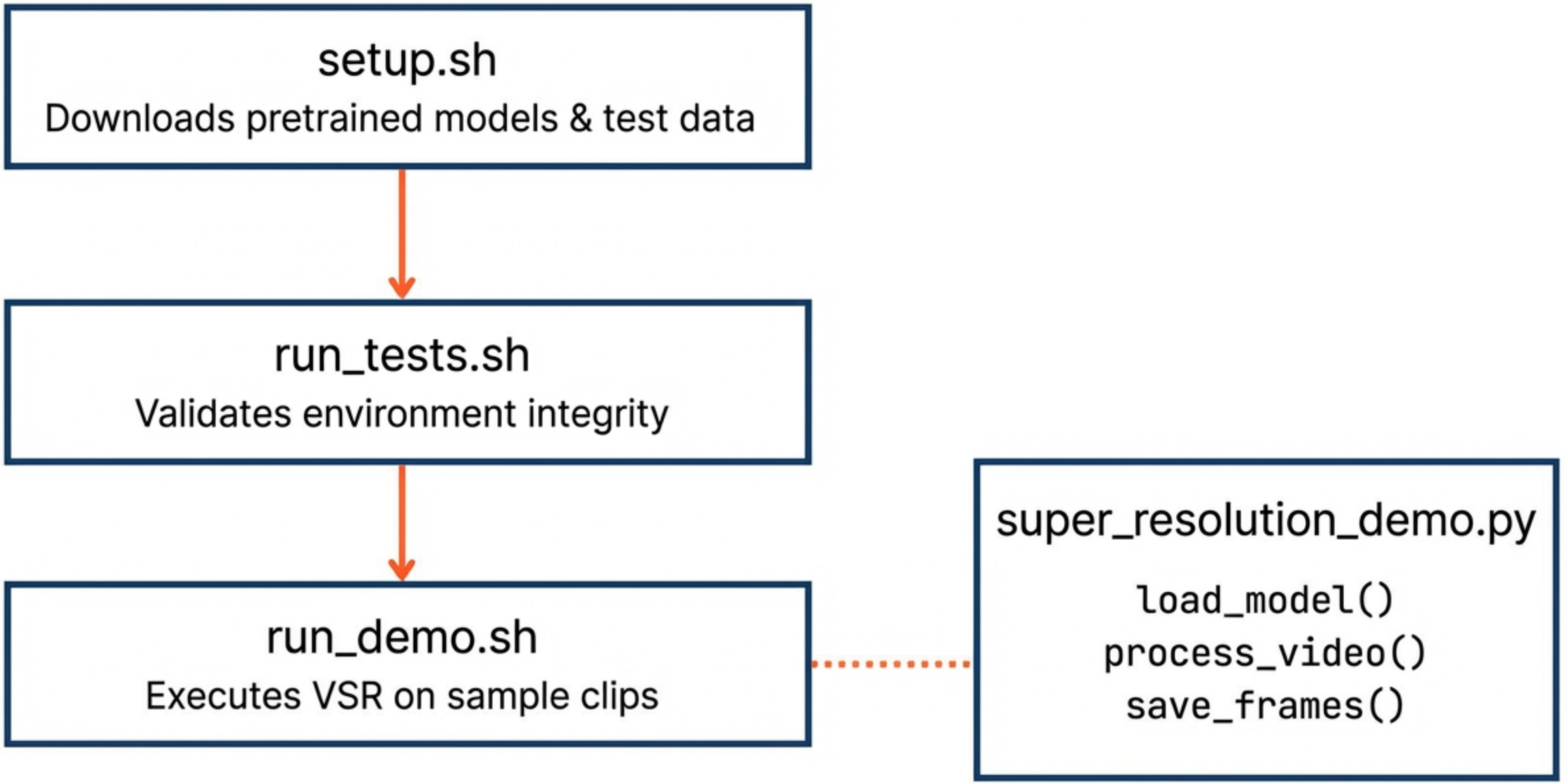
```
# Create virtual environment
python3 -m venv venv
source venv/bin/activate

# Install dependencies
pip install -r requirements.txt

# Check PyTorch version
python -c "import torch;
print(torch.__version__)"
```

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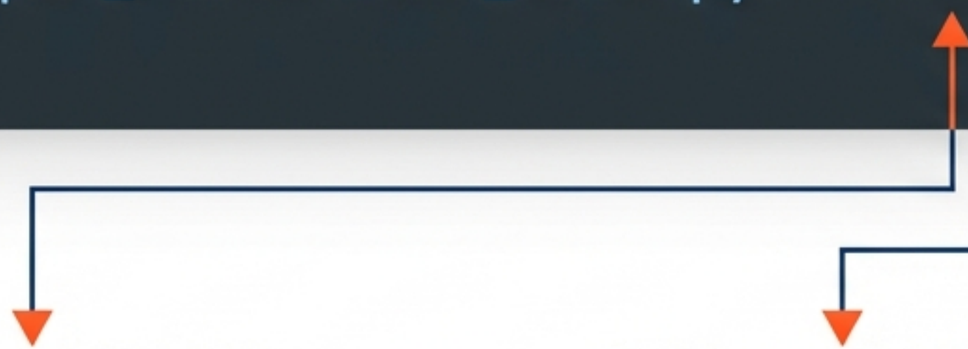
The Code Pipeline



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Running the Demo

```
python super_resolution_demo.py --scale 4 --device cuda --save
```



`--scale 4`: Upscaling factor (Options: 2, 4, 8). Higher scales require more VRAM.

`--device cuda`: Hardware acceleration. Essential for practical runtime.

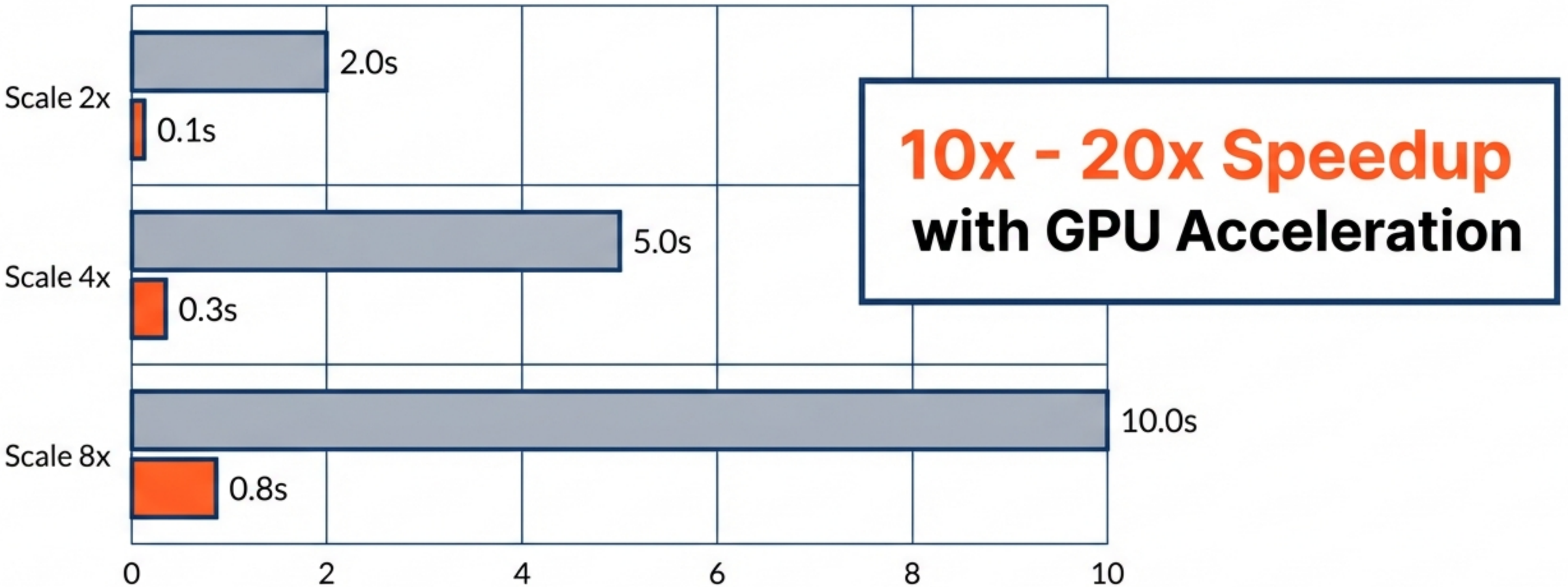
`--save`: Writes output frames to disk.

Pro Tip: Use `--device cpu`` only for debugging. CUDA is recommended for video processing.

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Processing Speed & Efficiency

Time per Image (in seconds)



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Evaluation Methodology



PSNR

Peak Signal-to-Noise Ratio

Measured in decibels (dB). Quantifies pixel-level reconstruction accuracy. Higher is better.



SSIM

Structural Similarity Index

Measures perceptual quality and structural integrity on a scale of 0 to 1. 1 is a perfect match.



Temporal Consistency

Frame Stability

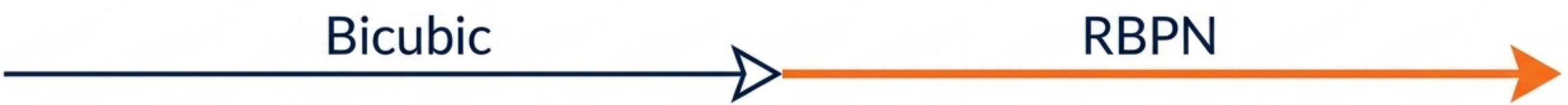
Measures jitter between consecutive frames. Lower difference values in static regions indicate better stability.

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Performance Benchmarks

Comparison: RBPN (Ours) vs. Bicubic Baseline

Dataset	Bicubic (Baseline)	RBPN (Ours)	Improvement
Vid4	27.13 dB	**31.40 dB**	+4.27 dB
Vimeo-90K	31.30 dB	**37.20 dB**	+5.90 dB
REDS	28.80 dB	**31.10 dB**	+2.30 dB



Significant quantitative improvement across all datasets.

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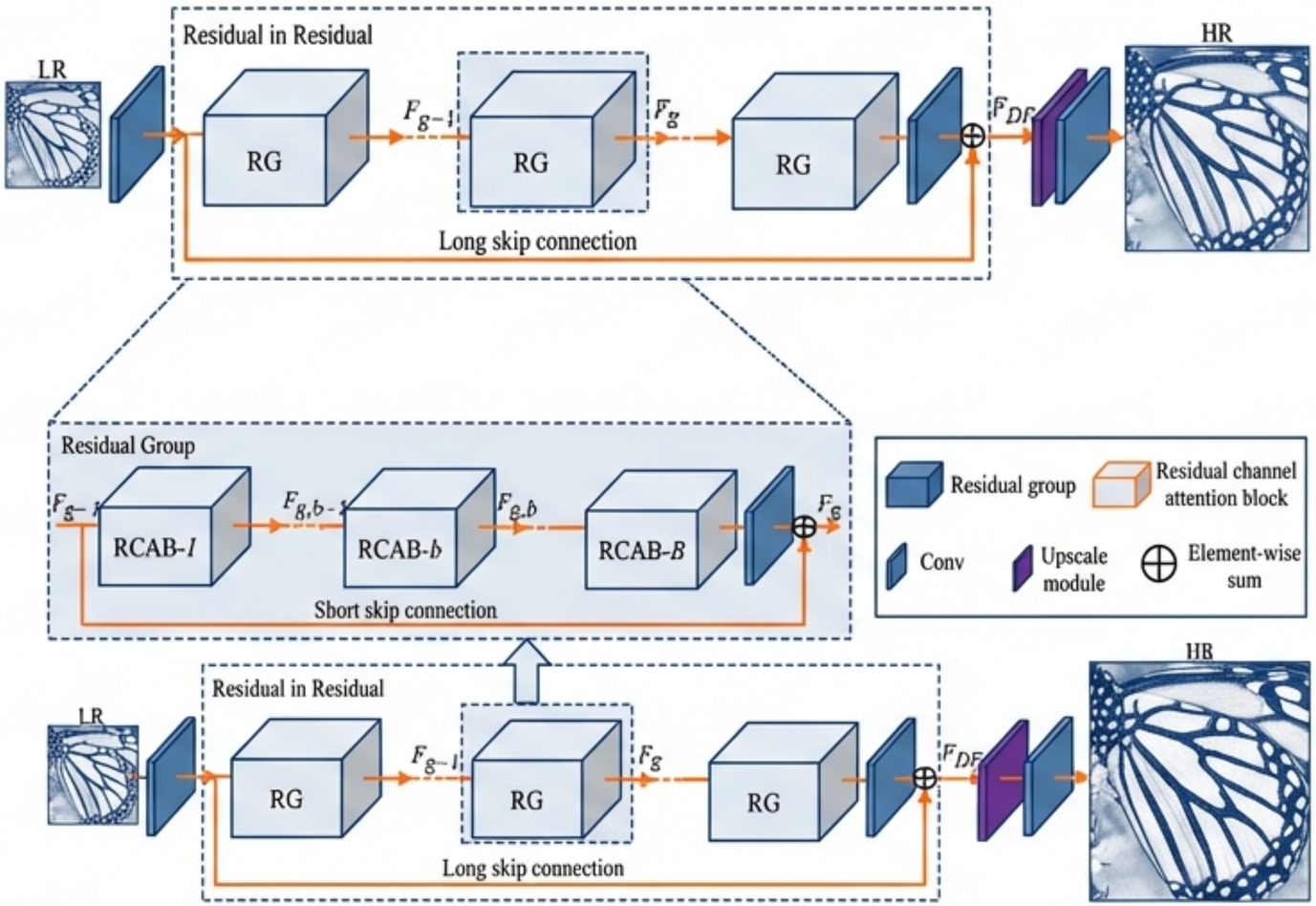
Comparison with State-of-the-Art

Competitors

- **EDVR**: Enhanced Deformable Convolutions.
- **BasicVSR**: Propagation-based efficiency.
- **RCAN**: Channel Attention Networks.

RBPN Strength

Explicit use of **projection error** for iterative refinement makes it highly effective at recovering fine textures compared to pure propagation methods.



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Challenges & Future Directions

Current Limitations

Computational Cost:
High memory usage (VRAM) at 8x scale.

Artifacts:
Potential ghosting in fast-motion scenes if optical flow fails.

Future Work

1. **Mobile NPU Optimization:**
   Porting models to Neural Processing Units for power efficiency.



2. **Real-time 4K:**
   Optimizing architecture for broadcast standards.



3. **ISP Integration:**
   Combining VSR with Electronic Image Stabilization (EIS).



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Conclusion

VSR leverages Time to enhance Space.

Method

RBPN integrates Back-Projection and Temporal Fusion.

Result

Significant gains (+4-5 dB) over baseline methods.

Application

Mobile zooming, streaming, and surveillance.

Try the Code: Clone the repository and run `run_demo.sh` today. https://github.com/JeffRen1977/super_resolution

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